

Heavy-Tailed Gradient Noise and Levy-Driven SGD

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Complexity of SGD in Finite Dimensional Space

Consider the unconstrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x),$$

where f has a Lipschitz continuous gradient, with constant $L > 0$, that is

$$\|\nabla f(y) - \nabla f(x)\|_* \leq L\|y - x\|, \quad x, y \in \mathbb{R}^n.$$

Applications:

- Neural Networks,
- Parameter estimation (MLE),
- Optimal control, etc.

Complexity of SGD in Finite Dimensional Space

Assuming that at every point $x \in \mathbb{R}^n$, we have access to stochastic gradient vectors:

$$g(x; \xi) \in \mathbb{R}^n,$$

that satisfy:

- **Unbiased estimator** $\mathbb{E}_{\xi}[g(x; \xi)] = \nabla f(x)$, and
- **Bounded variance** $\mathbb{E}_{\xi} [\|g(x; \xi) - \nabla f(x)\|_*^2] \leq \sigma^2$,

then to find a random point $\bar{x}$ using SGD such that $\mathbb{E}[\|\nabla f(\bar{x})\|_*] \leq \epsilon$, it is sufficient to perform

$$K = \mathcal{O} \left(L(f(x_o) - f^*) \cdot \left[\frac{1}{\epsilon^2} + \frac{\sigma^2}{\epsilon^4} \right] \right),$$

stochastic gradient steps. **Complexity of SGD is independent of the dimension n !**

SGD in Hilbert Space: Preliminaries

Let E and F be Banach spaces, and let $E_o \subset E$ be an open set, a function $f : E_o \rightarrow F$ is said to be Fréchet differentiable at $a \in E_o$ if there exists a bounded linear map $Df(a) \in \mathcal{L}(E, F)$ such that

$$\lim_{\substack{h \rightarrow 0 \\ a+h \in E_o}} \frac{\|f(a+h) - f(a) - Df(a)h\|_F}{\|h\|_E} = 0,$$

where $\mathcal{L}(E, F)$ is the space of bounded linear operators from E to F . If $E = H$ for some Hilbert space H , and $F = \mathbb{R}$, the operator $Df(a)$ is H -valued and we denote it as $\nabla f(a) \in H$.

SGD in Hilbert Space: Preliminaries

A linear operator $Q : H \rightarrow H$ is covariance operator of an H -valued random variable X if for all $\phi, \psi \in H$,

$$\langle Q\phi, \psi \rangle = \text{Cov}(\langle X, \phi \rangle, \langle X, \psi \rangle),$$

$\langle X, \phi \rangle$ is a scalar, so $\text{Cov}(\langle X, \phi \rangle, \langle X, \psi \rangle)$ is understood in the usual sense. If X has mean zero then,

$$\langle Q\phi, \psi \rangle = \text{Cov}(\langle X, \phi \rangle, \langle X, \psi \rangle) = \mathbb{E}[\langle X, \phi \rangle \langle X, \psi \rangle],$$

and $Q = \mathbb{E}[X \otimes X]$.

SGD in Hilbert Space: Preliminaries

In an infinite-dimensional Hilbert space H , **Parseval's identity** says that if $\{e_n\}_{n=1}^{\infty}$ is a complete orthonormal basis of H , then for every $x \in H$,

$$\|x\|^2 = \sum_{n=1}^{\infty} |\langle x, e_n \rangle|^2.$$

Equivalently,

$$x = \sum_{n=1}^{\infty} \langle x, e_n \rangle e_n$$

with convergence in the Hilbert-space norm.

The identity shown is

$$\langle \sigma(\phi)u, v \rangle = \langle u, \sigma(\phi)^*v \rangle.$$

This does extend to infinite-dimensional settings, but with an important condition.

When it is true. It holds in any Hilbert space provided that:

- $\sigma(\phi)$ is a bounded linear operator, and
- $\sigma(\phi)^*$ denotes its adjoint operator.

Consider the following optimization problem

$$\min_{\phi \in H} \mathcal{F}(\phi) := \mathbb{E}[\mathcal{F}_\gamma(\phi)],$$

where H is an infinite dimensional Hilbert space and $\{\mathcal{F}_\gamma : \gamma \in \Gamma\}$ is a family of functionals from H to $\mathbb{R}$, and γ is the source of uncertainty.

Gradient Descent for solving this problem,

$$\phi_{n+1} = \phi_n - \eta \nabla \mathcal{F}(\phi_n) = \phi_n - \eta \mathbb{E}[\nabla \mathcal{F}_\gamma(\phi_n)], \quad n \geq 0,$$

where $\eta > 0$ is the step-size of the learning and $\nabla \mathcal{F}$ is the Fréchet derivative of $\mathcal{F}$.

SGD in Hilbert Space

Computing $\nabla \mathcal{F}(\phi_n) = \mathbb{E}[\nabla \mathcal{F}_\gamma(\phi_n)]$ is expensive, motivating the use of Stochastic Gradients in the updates,

$$\phi_{n+1} = \phi_n - \eta \nabla \mathcal{F}_{\gamma_n}(\phi_n), \quad n \geq 0,$$

$\{\gamma_n\}_{n \geq 0}$ is a sequence of i.i.d random variables with the same distribution as γ . The SGD updates can be written in perturbation form as:

$$\phi_{n+1} = \underbrace{\phi_n - \eta \nabla \mathcal{F}(\phi_n)}_{\text{exact gradient step}} + \underbrace{\eta V(\phi_n; \gamma_n)}_{\text{noise}}, \quad n \geq 0, \quad (1)$$

where $V(\phi_n; \gamma_n) = \nabla \mathcal{F}(\phi_n) - \nabla \mathcal{F}_{\gamma_n}(\phi_n)$.

By construction, $\mathbb{E}[V(\phi_n; \gamma_n) \mid \sigma(\gamma_0, \dots, \gamma_{n-1})] = 0$, $n \geq 1$, with covariance operator $Q(\phi_n) = E_\gamma[V(\phi_n; \gamma_n) \otimes V(\phi_n; \gamma_n)]$.

SGD in Hilbert Space: stochastic modified equation (SME)

We are interested in studying the impact on the noise on gradient dynamics, to this end they propose the following continuous time model for analysis:

$$d\varphi_t = -\nabla \mathcal{F}(\varphi_t)dt + \sqrt{n} \sigma(\varphi_t)dB_t, \quad (2)$$

where B_t is a cylindrical Wiener process (Brownian motion on H), and σ is to be defined. The spaces of σ and B_t are to be determined to ensure consistency with SGD. Discretizing of SME using Euler-Maruyama, with step-size η (i.e., $t_k = k\eta$),

$$\tilde{\varphi}_{n+1} = \tilde{\varphi}_n - \eta \nabla \mathcal{F}(\tilde{\varphi}_n) + \sqrt{\eta} \sigma(\tilde{\varphi}_n)(B_{(n+1)\eta} - B_{n\eta}). \quad (3)$$

SGD in Hilbert Space

Comparing the SGD step (1) and the SME (2), the drift term matches the gradient. To match the covariance structure for any $u, v \in H$:

$$\begin{aligned} & \text{Cov}(\langle \sqrt{\eta} \sigma(\varphi)(B_{(n+1)\eta} - B_{n\eta}), u \rangle, \langle \sqrt{\eta} \sigma(\varphi)(B_{(n+1)\eta} - B_{n\eta}), v \rangle) \\ &= \eta \mathbb{E} \left[\langle B_{(n+1)\eta} - B_{n\eta}, \sigma(\varphi)^* u \rangle \langle B_{(n+1)\eta} - B_{n\eta}, \sigma(\varphi)^* v \rangle \right] \\ &= \eta^2 \langle \sigma(\phi) \sigma(\phi)^* u, v \rangle. \end{aligned}$$

For the discretization (3) to recover SGD, we need:

$$\text{Cov} \left(\sqrt{\eta} \sigma(\varphi)(B_{(n+1)\eta} - B_{n\eta}) \right) = \eta^2 \sigma(\phi) \sigma(\phi)^* = \eta^2 Q(\phi) = \text{Cov}(\eta V_n(\phi))$$

$\sigma : U \rightarrow H$ is the diffusion kernel of the noise process:

$$\sigma(\phi)u := \mathbb{E}_\gamma[V(\phi; \gamma)u(\gamma)], \quad u \in U$$

Let U and H be separable Hilbert spaces, let B_t be a Brownian motion on U , consider a general SME

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t, \quad X_0 = x \in H.$$

Theorem 2.8: Assume that $b : H \rightarrow H$ and $\sigma : H \rightarrow \mathcal{L}_2(U, H)$ are locally Lipschitz continuous and there exists $c > 0$ such that

$$\langle b(x), x \rangle_H + \|\sigma(x)\|_{\mathcal{L}_2(U, H)}^2 \leq c(1 + \|x\|^2), \quad x \in H.$$

Then for every $x \in H$ there exists a unique global adapted continuous solution X_t with $X_0 = x$.

SGD in Hilbert Space: Assumptions

The SME is well-posed if the following assumptions hold:

- ① : $\mathcal{F}$ is Fréchet differentiable on H .
- ② : The initial iterate ϕ_0 belongs to H .
- ③ : For every $\phi \in H$, the stochastic gradients have finite second moment

$$\mathbb{E}_\gamma [\|\nabla \mathcal{F}_\gamma(\phi)\|^2] < \infty.$$

- ④ : There exists random variables $C_L(\gamma) \in L^2(\Gamma)$ and $C_G(\gamma) \in L^1(\Gamma)$ such that for each $\phi, \psi \in H$,

$$\begin{aligned}\|\nabla \mathcal{F}_\gamma(\phi) - \nabla \mathcal{F}_\gamma(\psi)\| &\leq C_L(\gamma) \|\phi - \psi\|, \\ \langle \nabla \mathcal{F}_\gamma(\phi), \phi \rangle &\leq C_G(\gamma)(1 + \|\phi\|^2).\end{aligned}$$

What we are going to prove

If **A4**:

There exists random variables $C_L(\gamma) \in L^2(\Gamma)$ and $C_G(\gamma) \in L^1(\Gamma)$ such that for each $\phi, \psi \in H$,

$$\begin{aligned}\|\nabla \mathcal{F}_\gamma(\phi) - \nabla \mathcal{F}_\gamma(\psi)\| &\leq C_L(\gamma) \|\phi - \psi\|, \\ \langle \nabla \mathcal{F}_\gamma(\phi), \phi \rangle &\leq C_G(\gamma)(1 + \|\phi\|^2),\end{aligned}$$

then there exists $c > 0$ such that $\langle \nabla \mathcal{F}(\phi), \phi \rangle_H + \|\sigma(\phi)\|_{\mathcal{L}_2(U, H)}^2 \leq c(1 + \|\phi\|^2)$, which implies SME $d\varphi_t = -\nabla \mathcal{F}(\varphi_t)dt + \sqrt{n} \sigma(\varphi_t)dB_t$ is well-posed

It is sufficient to show the local Lipschitz continuity and linear growth for both $\nabla \mathcal{F}$ and σ .

Theorem 3.2: For every initial condition $\phi_0 \in H$, the SME (2) with the diffusion kernel $\sigma : U \rightarrow H$ defined as

$$\sigma(\phi)u := \mathbb{E}_\gamma[V(\phi; \gamma)u(\gamma)], \quad u \in U,$$

admits a unique adapted strong solution

$$\phi_t = \phi_0 - \int_0^t \nabla \mathcal{F}(\phi_s) ds + \sqrt{n} \int_0^t \sigma(\phi_s) dB(s), \quad t \geq 0,$$

such that

$$\phi \in L^p(\Omega; C([0, T]; H))$$

for every $p \geq 1$ and $T > 0$.

$$f(w) = \frac{1}{n} \sum_{i=1}^n f^{(i)}(w), \quad w^\star = \arg \min_{w \in \mathbb{R}^p} f(w).$$

SGD uses a minibatch Ω_k of size b :

$$w_{k+1} = w_k - \eta \nabla \tilde{f}_k(w_k), \quad \nabla \tilde{f}_k(w) = \frac{1}{b} \sum_{i \in \Omega_k} \nabla f^{(i)}(w).$$

Define stochastic gradient noise

$$U_k(w) = \nabla \tilde{f}_k(w) - \nabla f(w).$$

Brownian Model

The standard modeling assumption is

$$U_k(w) \sim \mathcal{N}(0, \sigma^2 I).$$

Then the SGD update can be written as

$$w_{k+1} = w_k - \eta \nabla f(w_k) + \sqrt{\eta} \sqrt{\eta \sigma^2} Z_k, \quad Z_k \sim \mathcal{N}(0, I).$$

For small η , this suggests the SDE

$$dw_t = -\nabla f(w_t) dt + \sqrt{\eta \sigma^2} dB_t.$$

Issues with the Brownian Model

- ① Measured gradient noise in deep nets has much heavier tails than Gaussian noise.
- ② Brownian paths are continuous, so escaping a basin usually means climbing the barrier.

For Brownian-driven small-noise dynamics, first exit times scale roughly like

$$\exp\left(\frac{2H}{\varepsilon^2}\right),$$

where H is a relevant barrier height.

That makes depth more important than width, which clashes with the folklore that SGD tends to prefer wide minima.

Gradient Noise Is Not Gaussian

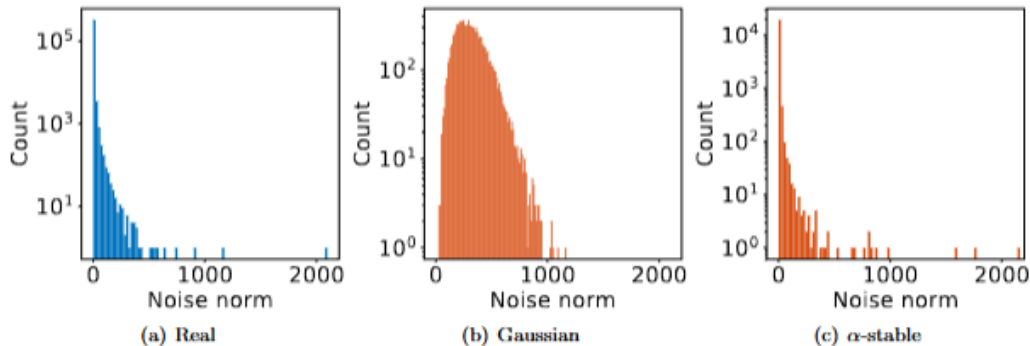


Figure 1: Comparison of Gradient Noise Norms

Stable Laws

The generalized central limit theorem says:

- finite variance sums $\Rightarrow$ Gaussian limit;
- heavy-tailed sums with infinite variance $\Rightarrow \alpha$ -stable limit.

The paper focuses on symmetric α -stable random variables:

$$X \sim S\alpha S(\sigma) \iff \mathbb{E}e^{i\omega X} = \exp(-|\sigma\omega|^\alpha).$$

Key facts:

- $\alpha \in (0, 2]$ is the tail index;
- smaller α means heavier tails;
- $\mathbb{E}|X|^r < \infty$ if and only if $r < \alpha$;
- $\alpha = 2$ recovers a Gaussian: $S2S(\sigma) = \mathcal{N}(0, 2\sigma^2)$.

From Stable Noise to Levy-Driven SDE

The paper assumes each coordinate of the gradient noise is stable:

$$[U_k(w)]_i \sim S\alpha S(\sigma(w)).$$

Then SGD can be rewritten as

$$w_{k+1} = w_k - \eta \nabla f(w_k) + \eta^{1/\alpha} (\eta^{(\alpha-1)/\alpha} \sigma) S_k,$$

where each coordinate of S_k has law $S\alpha S(1)$.

For small η , the continuous-time model becomes

$$dw_t = -\nabla f(w_t) dt + \eta^{(\alpha-1)/\alpha} \sigma dL_t^\alpha.$$

Here L_t^α is an α -stable Levy motion.

Definition of α -Stable Levy Motion

In the vector SDE above, each coordinate of L_t^α is an independent scalar α -stable Levy motion on $\mathbb{R}$. For $\alpha \in (0, 2]$, such a scalar process is determined by:

- **(i)** $L_0^\alpha = 0$ almost surely.
- **(ii)** For $t_0 < t_1 < \dots < t_N$, the increments $L_{t_i}^\alpha - L_{t_{i-1}}^\alpha$ are independent ($i = 1, \dots, N$).
- **(iii)** For $s < t$, the increment $L_t^\alpha - L_s^\alpha$ has the same law as L_{t-s}^α , namely $S\alpha S((t-s)^{1/\alpha})$.
- **(iv)** Stochastic continuity: for every $\delta > 0$ and $s \geq 0$, $\mathbb{P}(|L_t^\alpha - L_s^\alpha| > \delta) \rightarrow 0$ as $t \rightarrow s$.

When $\alpha = 2$, L_t^α agrees with $\sqrt{2} B_t$ for standard Brownian motion B_t .

Brownian Motion vs. Levy Motion

Feature	Brownian motion B_t	Stable Levy motion L_t^α
Increment law	Gaussian	α -stable
Tail behavior	light-tailed	heavy-tailed for $\alpha < 2$
Variance	finite	infinite for $\alpha < 2$
Paths	almost surely continuous	can have jumps
Basin escape	climb out	jump out

The crucial behavioral change is discontinuity: large jumps make barrier height less controlling.
 $\alpha = 2$ gives scaled Brownian motion.

Stable Densities and Levy Paths

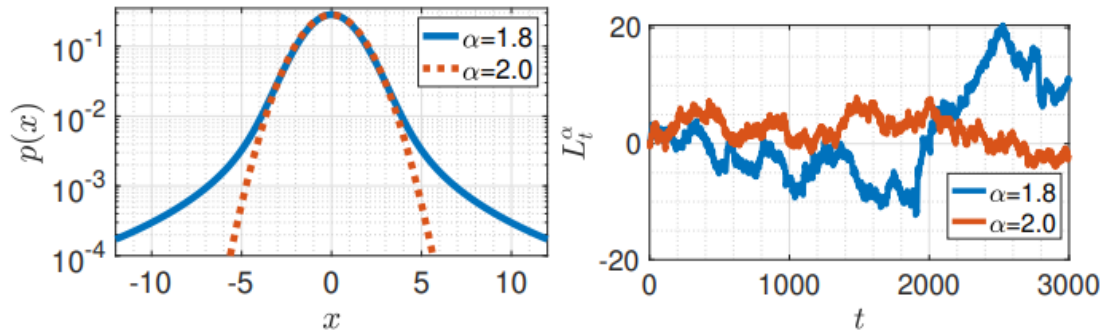


Figure 2: $S\alpha S$ Densities and Lévy Paths

Figure 2: Left - $S\alpha S$ densities; right - sample paths of L_t^α for $p = 1$. For $\alpha < 2$, the density has heavier tails and L_t^α displays jumps.

Metastability Setup

To isolate the basin-escape mechanism, the paper studies the 1D small-noise SDE:

$$dw_t^\varepsilon = -\nabla f(w_t^\varepsilon) dt + \varepsilon dL_t^\alpha.$$

Assume f has local minima $m_1, \dots, m_r$ separated by local maxima

$$-\infty = s_0 < m_1 < s_1 < \dots < s_{r-1} < m_r < s_r = \infty.$$

Each valley:

$$S_i = (s_{i-1}, s_i), \quad L_i = |s_i - s_{i-1}|.$$

Question: Starting near m_i , how long until the process reaches another basin?

Two Valleys

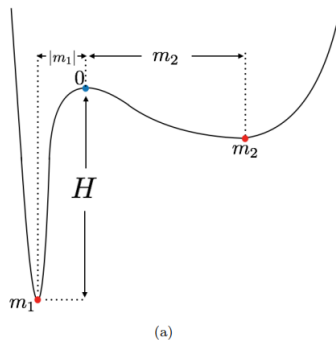


Figure 3: An objective with two local minima m_1 , m_2 separated by a local maximum at $s_1 = 0$.

Theorem 1: Levy Exit Times

For w_0 near m_i , as $\varepsilon \rightarrow 0$, transition times satisfy

$$\mathbb{P}_{w_0}(\varepsilon^\alpha T^i(\varepsilon) \geq u) \leq e^{-q_i u}.$$

The transition probabilities are controlled by

$$q_{ij} = \frac{1}{\alpha} \left| \frac{1}{|s_{j-1} - m_i|^\alpha} - \frac{1}{|s_j - m_i|^\alpha} \right|, \quad q_i = \sum_{j \neq i} q_{ij}.$$

Interpretation:

- transition time is polynomial in noise scale, on the order of $\varepsilon^{-\alpha}$;
- the rates depend on distances to valley boundaries;
- basin height does not appear.

Theorem 2: A Markov Chain Over Minima

After rescaling time by $\varepsilon^{-\alpha}$, the Levy-driven process converges to a continuous-time Markov chain on the minima:

$$w_{t\varepsilon^{-\alpha}}^\varepsilon \Rightarrow Y_{m_i}(t), \quad Y(t) \in \{m_1, \dots, m_r\}.$$

The generator has off-diagonal entries

$$q_{ij} = \frac{1}{\alpha} \left| \frac{1}{|s_{j-1} - m_i|^\alpha} - \frac{1}{|s_j - m_i|^\alpha} \right|,$$

and diagonal entries

$$q_{ii} = - \sum_{j \neq i} q_{ij}.$$

The stationary probabilities solve

$$Q^T \pi = 0.$$

Wide Valleys in the Two-Minima Case

For two minima with $m_1 < 0 < m_2$ and the barrier at $s_1 = 0$, the paper gives

$$\pi_1 = \frac{|m_1|^\alpha}{|m_1|^\alpha + |m_2|^\alpha}, \quad \pi_2 = \frac{|m_2|^\alpha}{|m_1|^\alpha + |m_2|^\alpha}.$$

So if $|m_2| > |m_1|$, the second valley is wider and

$$\pi_2 > \pi_1, \quad \frac{\pi_2}{\pi_1} = \left(\frac{|m_2|}{|m_1|} \right)^\alpha.$$

Estimating the Tail Index

The paper uses an estimator specialized for stable laws.

Let $X_1, \dots, X_K \sim S\alpha S(\sigma)$ and $K = K_1 K_2$. Define block sums

$$Y_i = \sum_{j=1}^{K_1} X_{j+(i-1)K_1}, \quad i = 1, \dots, K_2.$$

The estimator is

$$\widehat{\frac{1}{\alpha}} = \frac{1}{\log K_1} \left(\frac{1}{K_2} \sum_{i=1}^{K_2} \log |Y_i| - \frac{1}{K} \sum_{i=1}^K \log |X_i| \right).$$

Then $\widehat{1/\alpha} \rightarrow 1/\alpha$ almost surely as $K_2 \rightarrow \infty$.

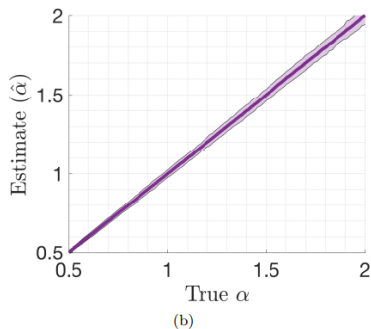


Figure 4: Tail-index estimator illustration.

Experimental Protocol

At selected SGD iterations, the authors:

- 1 compute the full gradient $\nabla f(w_k)$;
- 2 partition the data into disjoint minibatches of size b ;
- 3 compute minibatch gradients $\nabla \tilde{f}_{\Omega_k^i}(w_k)$;
- 4 form gradient noises

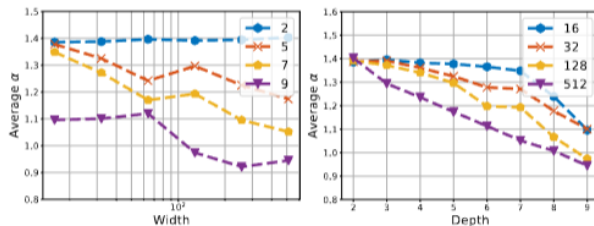
$$U_k^i(w_k) = \nabla \tilde{f}_{\Omega_k^i}(w_k) - \nabla f(w_k);$$

- 5 concatenate coordinates and estimate α .

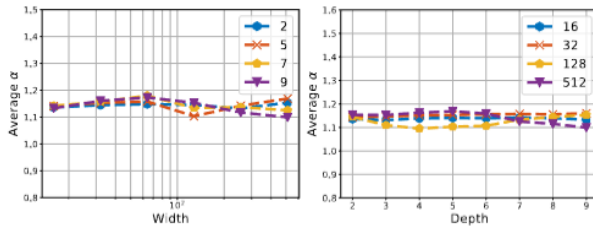
Experiments vary:

- architecture: fully connected nets and AlexNet-like CNNs;
- dataset: MNIST, CIFAR10, CIFAR100;
- network size, minibatch size, loss function, and random seed.

Results: Network Size - FCN Width and Depth

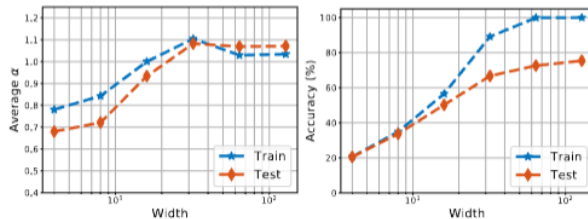


(a) MNIST

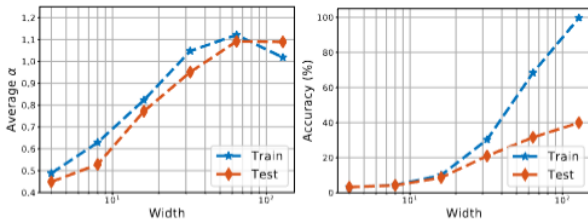


(b) CIFAR10

Results: Network Size - CNN Width

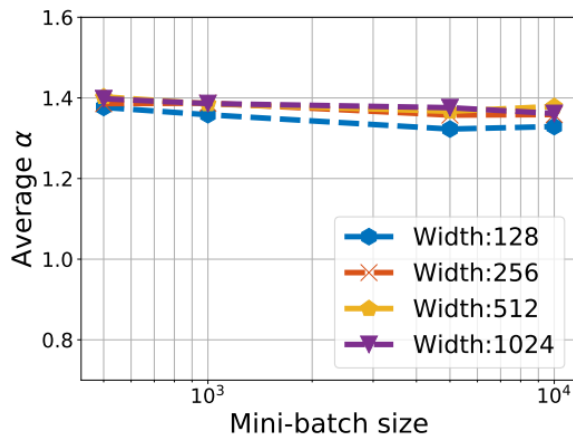


(a) CIFAR10

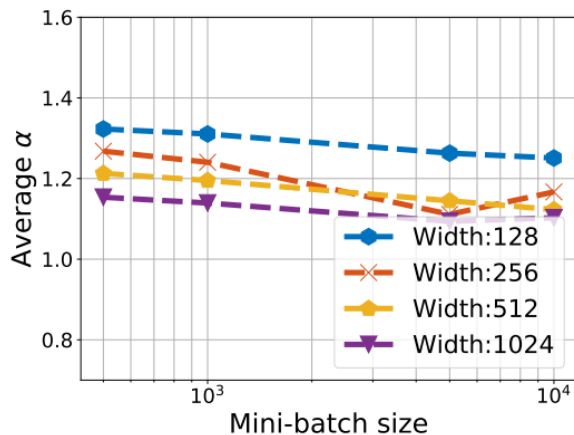


(b) CIFAR100

Results: Minibatch Size



(a) Depth = 2



(b) Depth = 4

Brownian Results vs. Levy Results

Question	Brownian model	Levy model
Noise law	Gaussian	α -stable
CLT justification	finite variance CLT	generalized CLT
Continuous-time driver	B_t	L_t^α
Path geometry	continuous	jumps for $\alpha < 2$
Escape scaling	exponential in barrier height	polynomial in ε^{-1}
Wide minima story	hard to explain locally	supported by metastability

Limitations and Open Problems

- The dependence on learning rate and minibatch size is not fully characterized.
- The tail index may change with the current state w_k and time.
- Deep-network landscapes have many nearly flat directions, while the clean theory assumes nondegenerate minima.
- Empirical heavy tails do not by themselves prove that an exact stable law governs every coordinate.

Closing Summary

heavy-tailed gradient noise $\Rightarrow$ α -stable model $\Rightarrow$ Levy-driven SDE $\Rightarrow$ jumps $\Rightarrow$ width-dependent